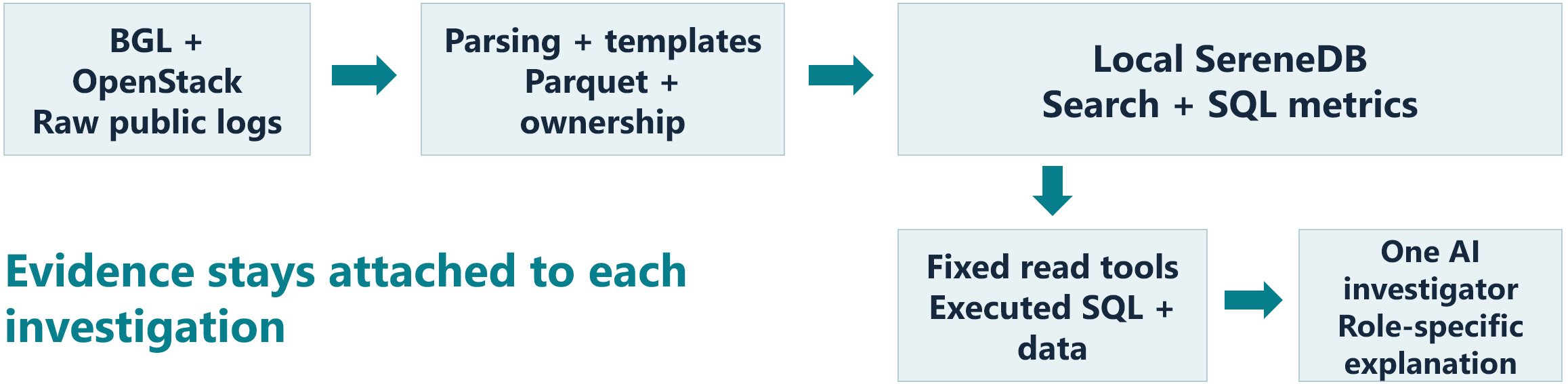
Log Observatory

A supported next step for every operations role

Nearly 5 million real log lines. Search, SQL evidence and an AI investigator.

Engineer / Team Lead / Manager

SereneDB calculates. The agent explains



Evidence stays attached to each investigation

Streamlit for the team
MCP for compatible assistants

Observed facts, hypotheses and next checks remain separate.

Self-built parsing, schema, SQL loader and fictional ownership. SereneDB runs locally in Docker.

Engineer: the exception is inside the traceback

8

matching failure headers

InvalidDiskInfo

qemu-img cannot open the instance disk:
“No such file or directory”

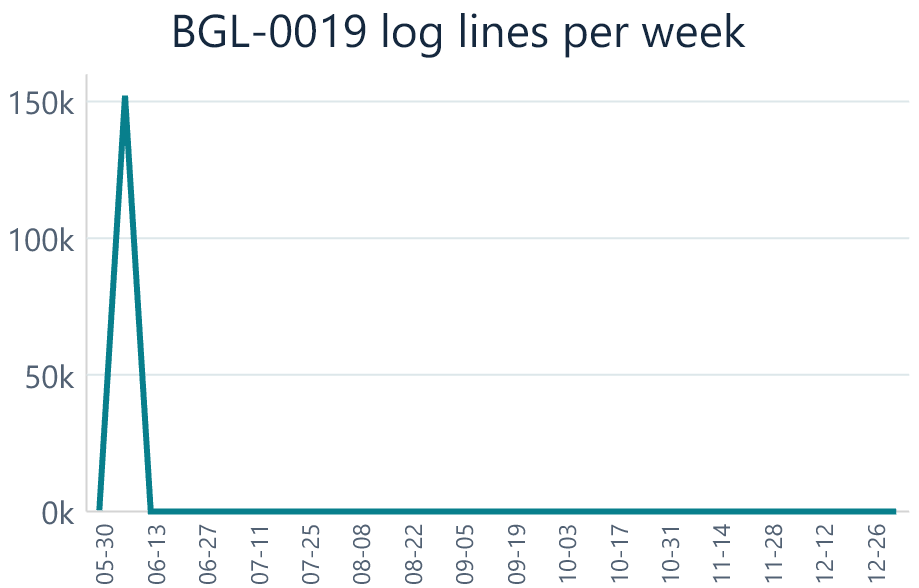
Next check

Inspect the disk path and correlate VM deletion or storage activity.

The inspected traceback establishes the immediate failure.
The underlying reason and the other seven reports still need checking.

Team Lead: recurring patterns and their reach

Pattern	Log lines	Nodes
TLB error	152,734	541
Storage interrupt	63,491	22,529
Instruction address	47,563	22,791

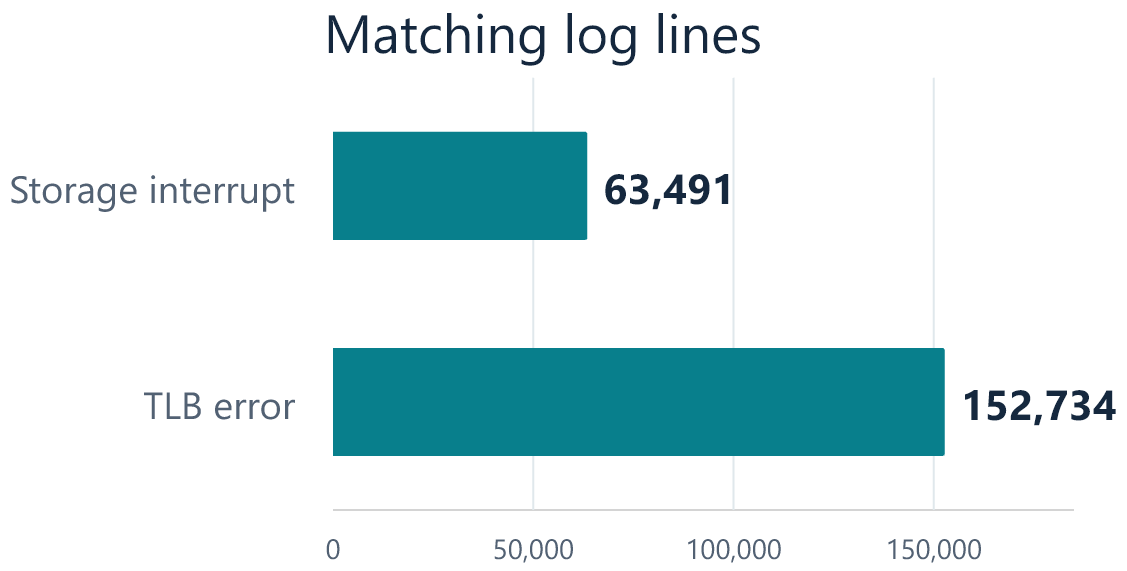


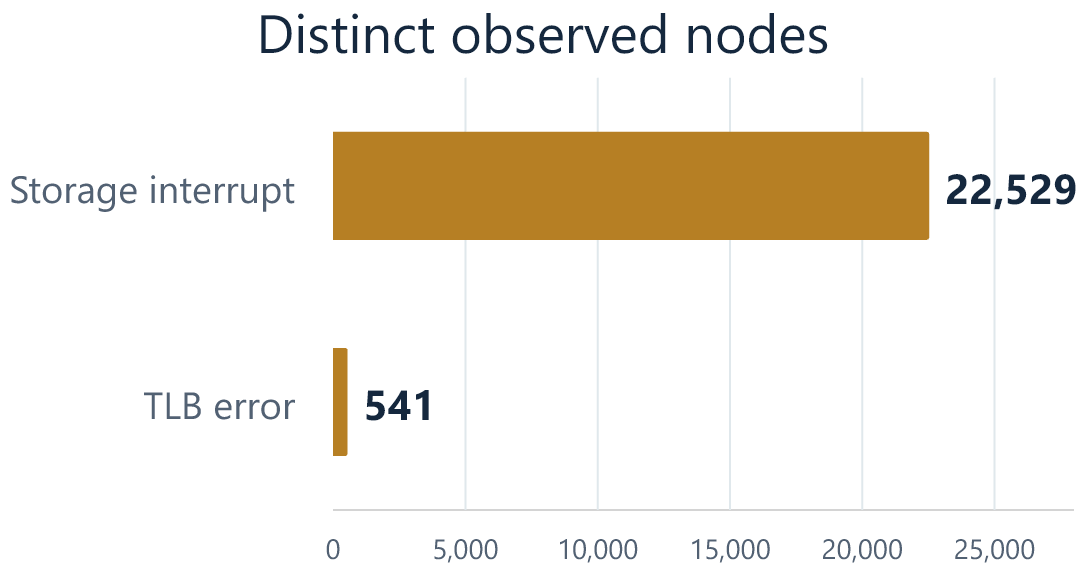
152,182 TLB lines in one weekly bucket

Check the localized burst and the wider storage pattern.
Log lines and observed nodes do not establish customer impact.

HPC Kernel / BGL ERROR + FATAL / historical full period / top_patterns and pattern_trend

Manager: frequency and reach change the priority





Storage interrupts rank first in the BGL node comparison

Score factors: frequency, severity, reach, trend and BGL alerts.
The score supports investigation and does not measure economic loss.

BGL / all teams, full recorded period / exact-phrase full-match aggregations and priority_ranking

The same tools work through MCP

7 read tools

3 role prompts

Bounded results
Original SQL evidence resources
No model calls inside the server

corpus_overview
search_logs
get_context
vm_dauer
top_patterns
pattern_trend
priority_ranking

Claude Desktop / Claude Code setup included

MCP SDK checks passed in both protocol modes. Actual Claude answer recording remains a team acceptance step.

A supervised pilot with an operations team

Less manual log lookup.
Evidence that survives a handover.

Pilot step	What we would measure
Connect one customer log source	Time to a supported next action
Compare with the current workflow	Answer quality and evidence accuracy
Extend format and ownership adapters	Investigation cost and operator feedback

Proposed pilot and expected benefits. Customer time savings and return on investment have not been measured.

Evidence first, human decisions

Working today

4,955,783 historical log lines
Search and aggregation with SQL evidence
Three role-specific agent workflows

Limits and next step

Fictional ownership. Heuristic events and scores.
Human verification before action.
Next: a customer pilot with access controls and sensitive-data handling.

Appendix: discoverable tool design

Role / tool	Key inputs	Result
All: corpus_overview	none	Dataset / team counts and dates
Engineer: search_logs	query, dataset, team, dates, limit	Matching logs + full-count statistics
Engineer: get_context	log_id, window, limit	Request / VM / node event context
Engineer: vm_dauer	instance_id, limit	Spawn observation + SQL baseline
Lead: top_patterns	team, dataset, dates, limit	Template counts, events and reach
Lead / Manager: pattern_trend	template / team, interval, dates	Time buckets + observation gaps
Manager: priority_ranking	group_by, dataset, team, dates	Score + individual factors

Inputs are validated. Outputs retain executed SQL, data, scope and explicit truncation. No arbitrary SQL tool.

Appendix: executed SQL and supporting results

```
SELECT template_id, COUNT(*) AS log_lines,
       COUNT(DISTINCT NULLIF(NULLIF(node, 'NULL'), '-')) AS nodes
FROM logs
WHERE dataset = 'BGL' AND level IN ('ERROR', 'FATAL')
  AND message IN ('data TLB error interrupt', 'data storage interrupt')
GROUP BY template_id
ORDER BY log_lines DESC
```

Template	Matching log lines	Distinct observed nodes
BGL-0019	152,734	541
BGL-0119	63,491	22,529

Observed nodes describe dataset reach, not measured customers or outage cost.

Actual read query executed on the local full BGL corpus on 18 Oct 2026. Full evidence is included with the submission.